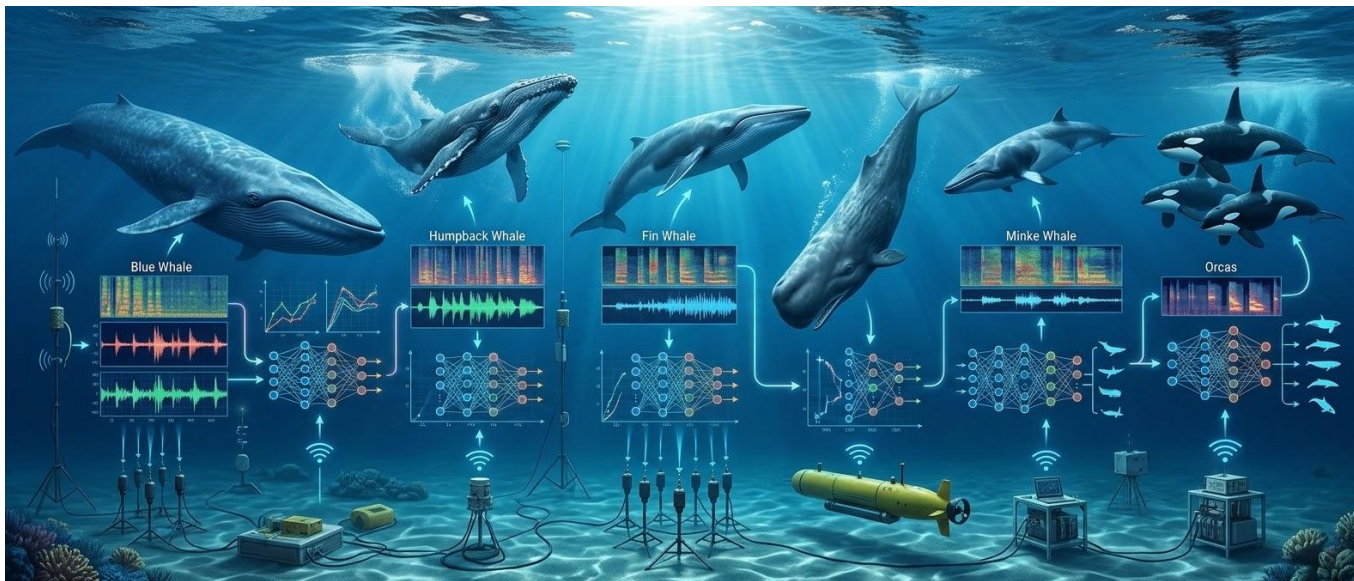


Deep Learning in the Shallow End

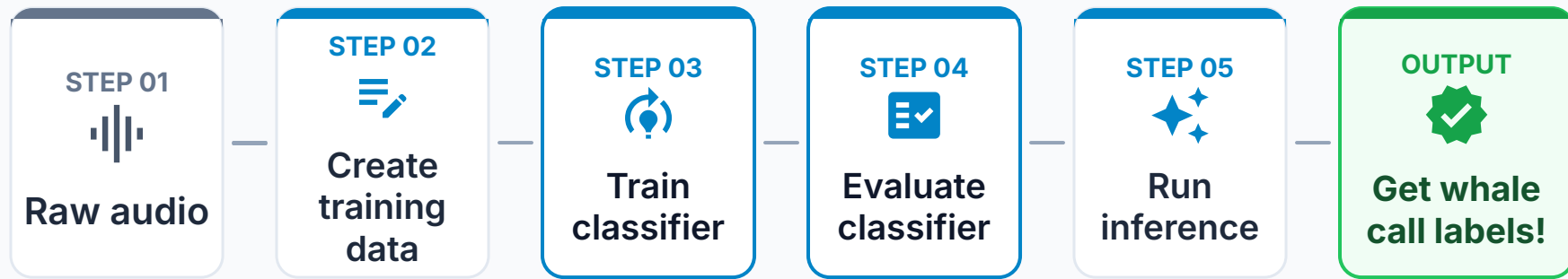
Accelerating PAM Analysis by lowering the bar to creating whale-tuned machine learning models

Ann Allen, Dan Woodrich, Shannon Rankin

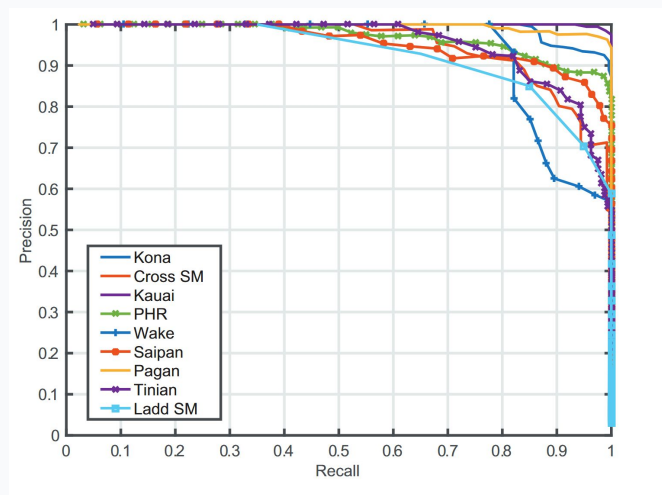


(Models that are way better than this AI image)

Traditional ML Workflow

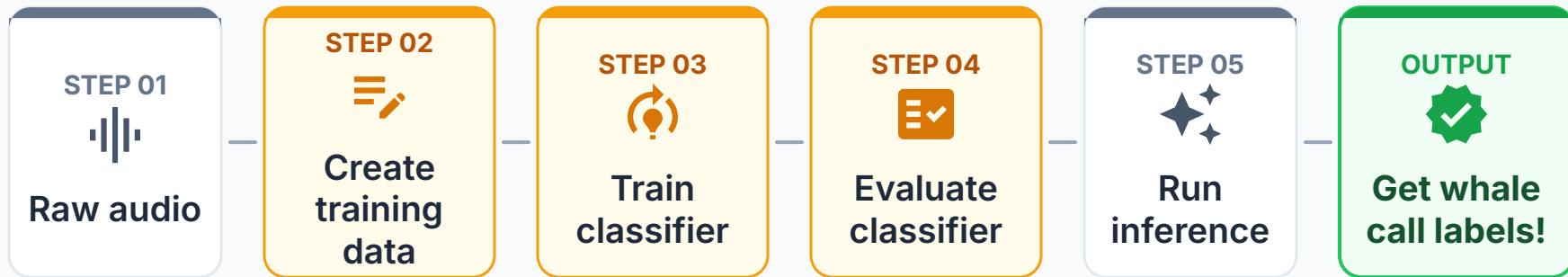


Performance: Machine Learning (ML) models do a very good job of learning to recognize whale calls



Traditional ML Workflow

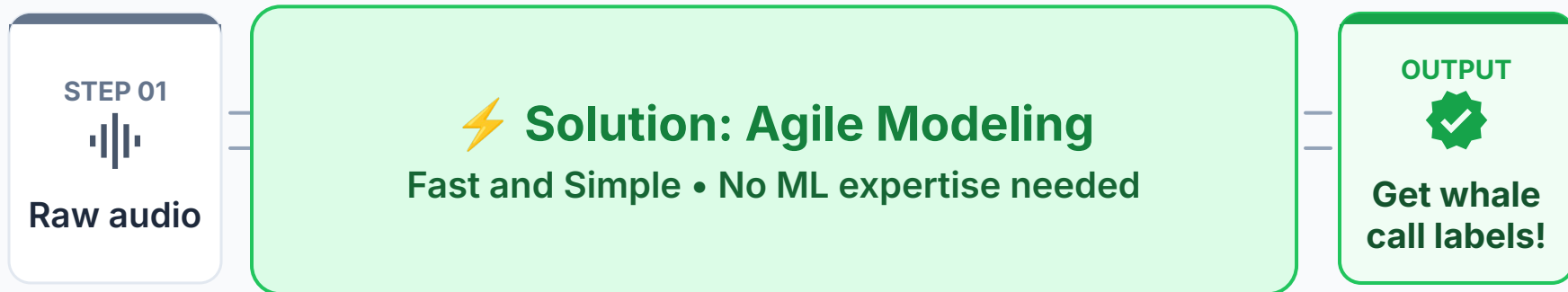
⚠ Bottleneck: Slow and analyst/expert intensive process



⚠ Challenges with Traditional Bioacoustics ML Workflow:

- **High Data Requirement:** Requires hundreds to thousands of labeled calls and several rounds of training.
- **Time Consuming:** Annotation and iterative training cycles take significant analyst hours.
- **Expert Dependency:** Requires heavy involvement and availability of ML experts.
- **Does Not Scale:** Building classifiers requires re-training from scratch for every new call type or dataset.

Agile Modeling Workflow



✦ Agile Modeling Definition & Process Advantages:

- **Definition:** Agile modeling is a process for rapidly developing classifiers using embeddings from a pre-trained 'foundation' model.
- **Fast & Simple:** Streamlines iterations and eliminates waiting on complex ML training pipelines.
- **Accessibility:** No ML expertise needed—empowers domain experts and scientists directly.
- **Efficient Active Learning:** Easily evaluate, search embeddings, and retrain small classifiers on minimal examples.

Perch Hoplite Workflow



User Friendly GUI

INPUT



Raw Audio

Unlabeled
acoustic
recordings

STEP 01

Generate Embeddings

Pre-trained AI foundation models process audio to extract compact, rich vector representations.

 **Slow & Computationally Expensive**

BUT it is done **only 1x** and the output vectors are saved to cloud database.

STEP 02

Few-Shot Active Learning

Classifier is trained to recognize new sounds using only a handful of examples, rather than thousands, by utilizing pre-trained embeddings.

 **FAST & Easily Modifiable**

Done rapidly for each new call type.

OUTPUT



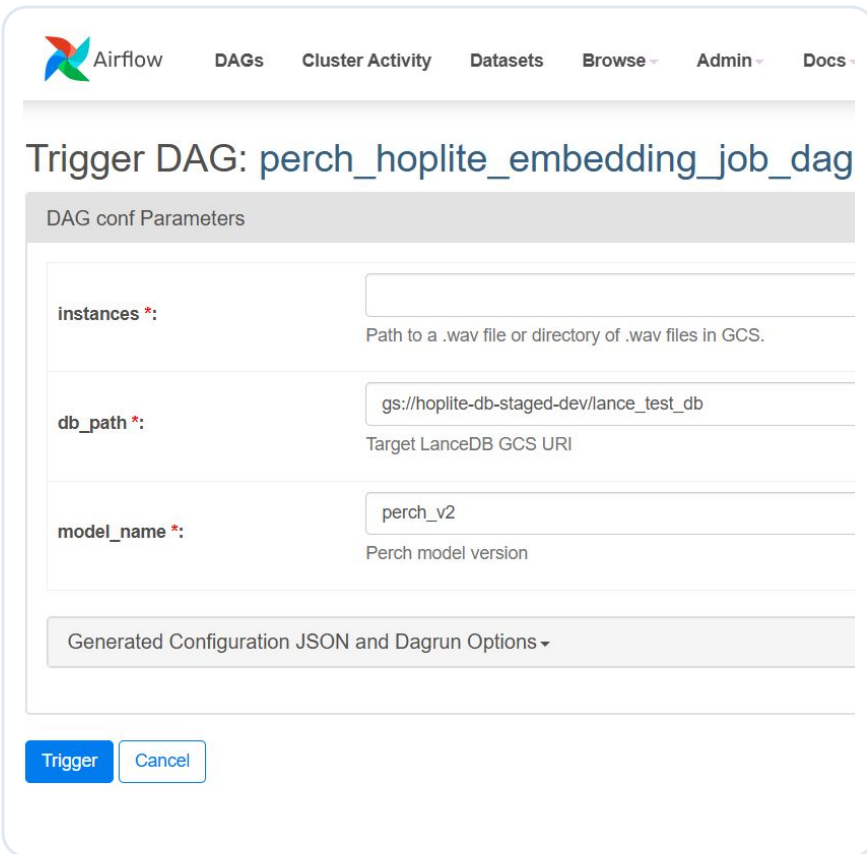
**Get Whale
Call Labels!**

Step 1: Embeddings, the user experience

Embeddings Generation in Apache Airflow DAG

You input:

- Audio location: Run on an individual file or an entire directory of audio files
- Embeddings db name: Generated vectors are automatically saved in a LanceDB database
- Model Selection: Choose which foundation model to use for embedding extraction. Default will be perch_v2



The screenshot displays the Apache Airflow web interface. At the top, there is a navigation bar with the Airflow logo and links for DAGs, Cluster Activity, Datasets, Browse, Admin, and Docs. The main heading is "Trigger DAG: perch_hoplite_embedding_job_dag". Below this, a section titled "DAG conf Parameters" contains three input fields: "instances *" (empty), "db_path *" (containing "gs://hoplite-db-staged-dev/lance_test_db"), and "model_name *" (containing "perch_v2"). Each field has a descriptive label to its right. At the bottom of the configuration section, there is a button labeled "Generated Configuration JSON and Dagrun Options". Below the configuration section, there are two buttons: "Trigger" (in blue) and "Cancel" (in white with a blue border).

Airflow DAGs Cluster Activity Datasets Browse Admin Docs

Trigger DAG: perch_hoplite_embedding_job_dag

DAG conf Parameters

instances *:
Path to a .wav file or directory of .wav files in GCS.

db_path *:
gs://hoplite-db-staged-dev/lance_test_db
Target LanceDB GCS URI

model_name *:
perch_v2
Perch model version

Generated Configuration JSON and Dagrun Options

Trigger Cancel

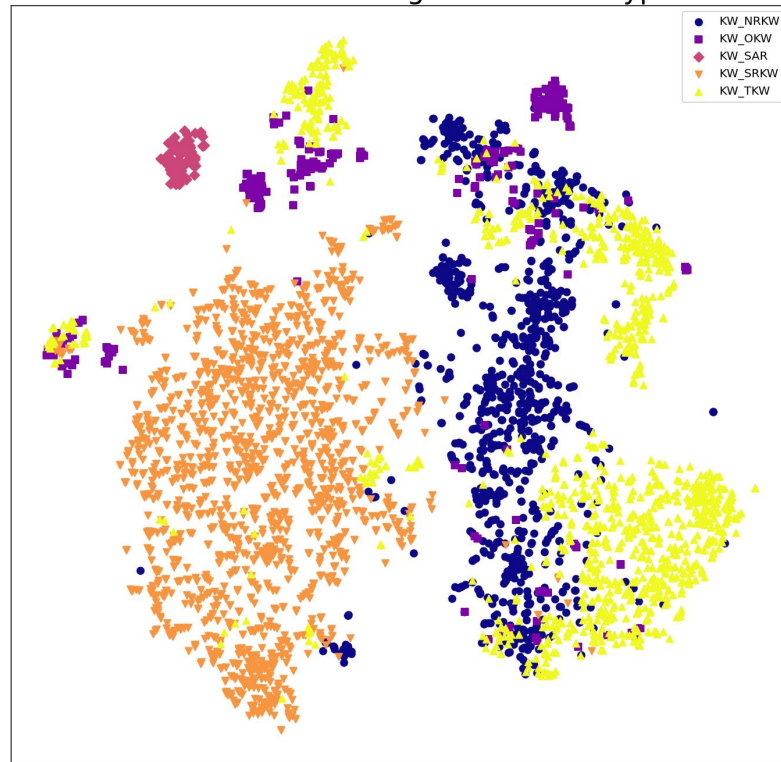
Step 1: Embeddings, the details

Raw audio passes through a frozen foundational ML model to learn general features of sound (pitch, harmonics, rhythm), translating complex audio into a dense mathematical vector—like a 'fingerprint' of 5-second clips.

- **Model Flexibility:** Any ML model can be used, but performance will vary.
- **Perch 2.0:** Pre-trained on a massive, diverse audio dataset for bird calls.
- **One-Time Execution:** This step is slow and computationally intense BUT it is done only **ONCE**, then vectors are saved to a cloud database.

tSNE Plot

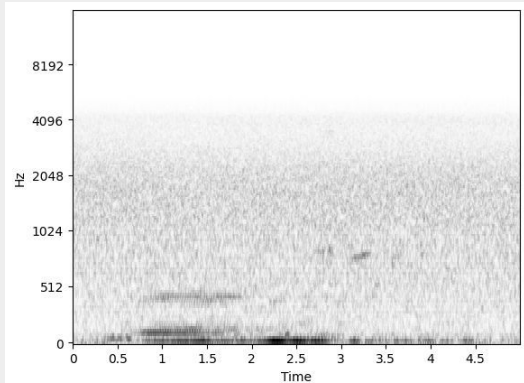
Perch 2.0 Embeddings: DCLDE Ecotype



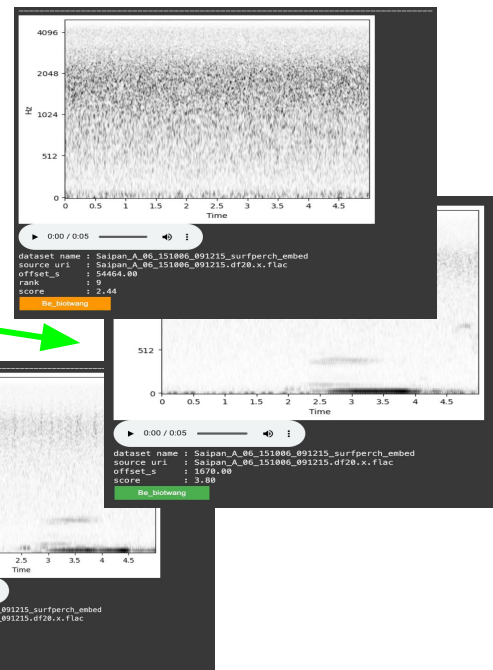
Step 2: Few-Shot Active Learning, the user experience

1. Provide sound examples

Provide a few examples of a new sound



2. Evaluate output of similar sound results



Step 2: Few-Shot Active Learning, the details

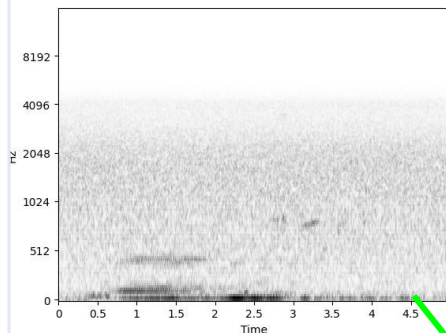
Sound Examples: Analyst provides a handful of examples of new target sound

Train Classifier:

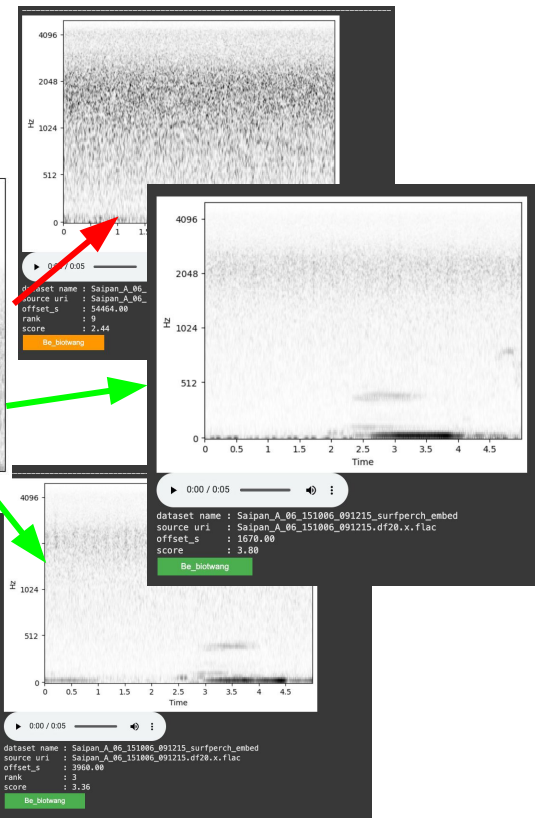
- Hoplite rapidly searches embeddings for similar examples
- Analyst provides feedback on the results
- Hoplite quickly trains a custom classifier based on feedback
- Analyst provides more feedback on the results of the classifier

Evaluate and Scale: The refined classifier is applied to larger dataset to label all data

Sound Example



Similar Sound Results



Images: Lauren Harrell

Annotation Database

Google Cloud Services

- File start times & durations
- Internal structure: channels, subchunks (xwav)

Makara

- Data Provenance: Deployments, recorders, locations
- Species detection summaries

Annotation DB

- Individual call annotations & effort for both manual and machine learning efforts
- *Highly optimized for fast queries at scale (millions of annotations)*
- GCS acoustic file metadata

Admin app

Manages and validates metadata

Big Query?

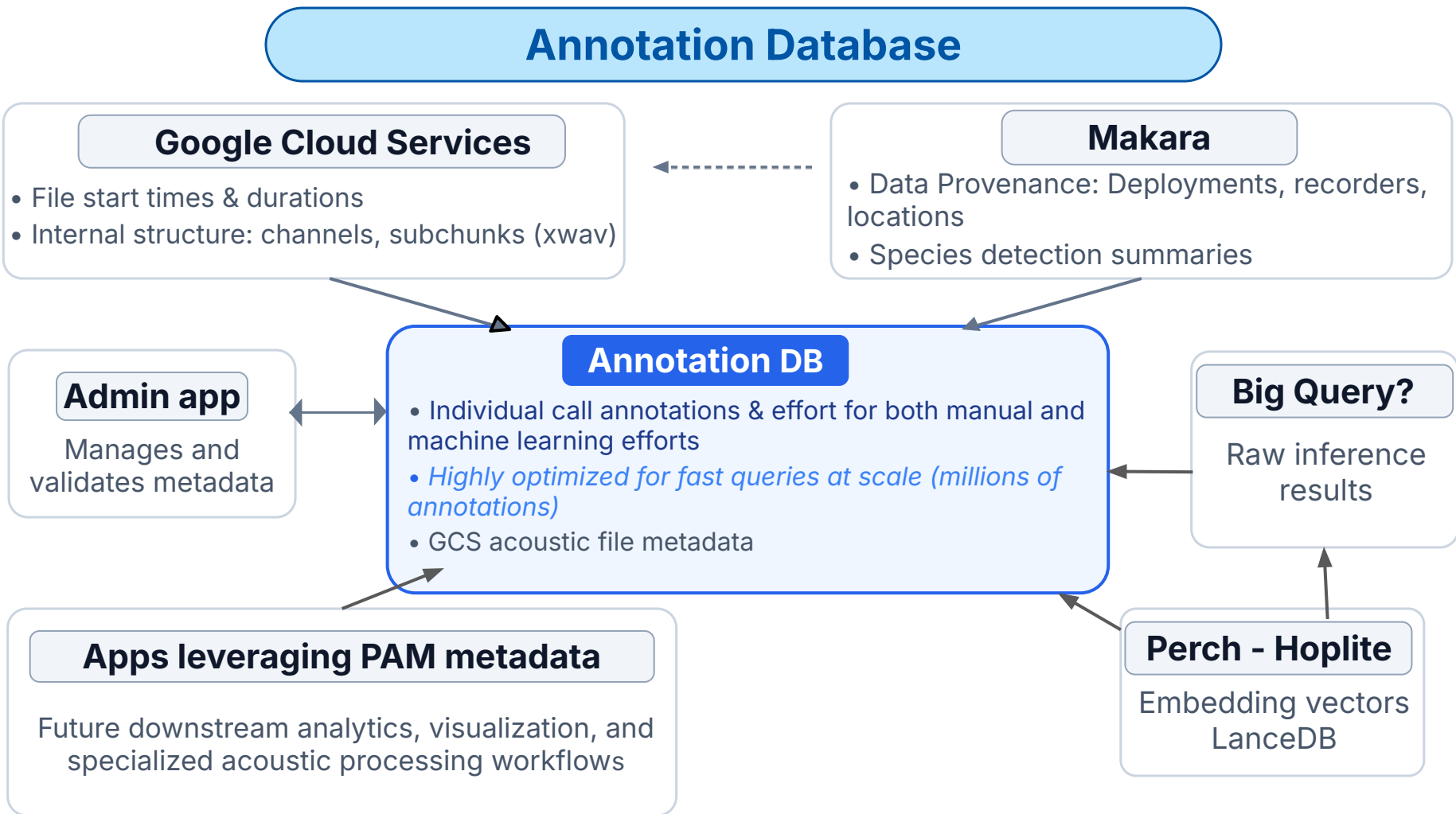
Raw inference results

Apps leveraging PAM metadata

Future downstream analytics, visualization, and specialized acoustic processing workflows

Perch - Hoplite

Embedding vectors
LanceDB



The Fin-ish

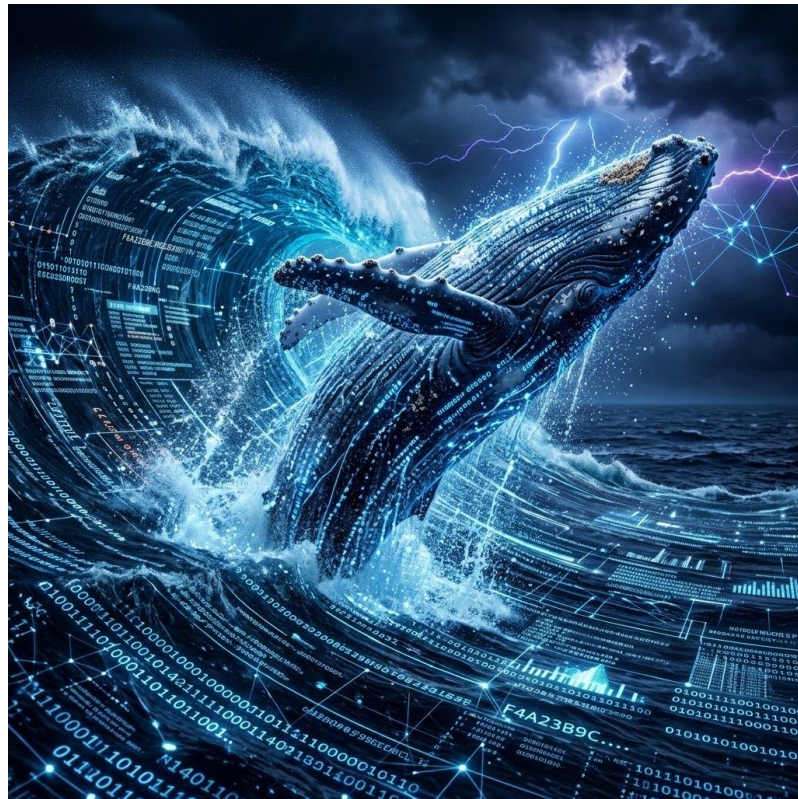
Project Goal

To make machine learning accessible so you don't need an AI background to dive right in, your team can swiftly navigate new call annotations and smoothly drift right into the downstream analysis.

Want to be a beta tester and get ahead of the wave?

BReach out!

We need examples of the different data types collected across science centers.



Resources

CODE & DEMO

Perch Hoplite Whale Demo

https://github.com/google-research/perch/blob/main/chirp/projects/whale_demo/agile_modeling_noaa_demo.ipynb

REPOSITORY

Perch Hoplite GitHub

<https://github.com/google-research/perch-hoplite>

ARXIV PAPER

Perch Bird Model Transfers to Whale

<https://arxiv.org/html/2512.03219v1>

ARXIV PAPER

Agile Modeling arXiv Paper

<https://arxiv.org/abs/2505.03071>

Acknowledgements

Special thanks to the Google Team, including Lauren Harrell, and the TSPI team, including Jordan Shimer and Jonathan Windmiller.